

P48.10

$$\text{已知 } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 ; \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

$$(1) \text{ 设 } p(x, y) = u + v, \quad \frac{\partial^2 p}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} \right) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x^2}$$

$$\therefore \frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 v}{\partial y^2} = 0 \quad \square$$

$$(2) \text{ 设 } q(x, y) = u \cdot v$$

$$\frac{\partial^2 q}{\partial x^2} = \frac{\partial}{\partial x} \left(v \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial x} \right) = \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} + v \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + u \frac{\partial^2 v}{\partial x^2}$$

$$\text{同理 } \frac{\partial^2 q}{\partial y^2} ; \text{ 有 } \frac{\partial^2 q}{\partial x^2} + \frac{\partial^2 q}{\partial y^2} = v \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + u \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + 2 \left(\frac{\partial v}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} \right)$$

$$= 2 \left(\frac{\partial v}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} \right) \neq 0 \quad \text{不是}$$

P48 14.(2)(3)

$$(2) \begin{cases} \frac{\partial v}{\partial x} = 4y = -\frac{\partial u}{\partial y} & \therefore u = -2y^2 + h(x) \\ \frac{\partial v}{\partial y} = 4x = \frac{\partial u}{\partial x} & \frac{\partial u}{\partial x} = h'(x) = 4x \Rightarrow h(x) = 2x^2 + C \end{cases}$$

$$\Rightarrow u = -2y^2 + 2x^2 + C$$

$$\text{由 } u(0) = 1 = C \quad \therefore C = 1, \quad u = -2y^2 + 2x^2 + 1$$

$$\Rightarrow f(z) = -2y^2 + 2x^2 + 1 + i \cdot 4xy = 2z^2 + 1$$

$$(3) \begin{cases} \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -e^x \cos y & v = -e^x \cos y + h(y) \\ \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = e^x \sin y & \frac{\partial v}{\partial y} = e^x \sin y + h'(y) \therefore h'(y) = 0 \end{cases}$$

$$v = -e^x \cos y + C$$

$$\therefore f(z) = e^x \sin y + i(-e^x \cos y + C) = -e^{iz} + iC$$

17. (1)(4)

$$(1) i+1 = e^{i(\frac{\pi}{2}+2k\pi)}(1+i) = e^{i(\frac{\pi}{2}+2k\pi)+i(\frac{\pi}{4}+2k\pi)}$$

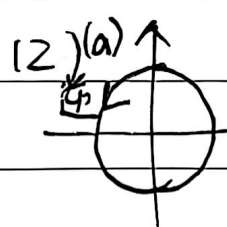
$$= e^{i(\frac{3\pi}{4}+2k\pi)} [\cos(\frac{3\pi}{4}+2k\pi) + i \sin(\frac{3\pi}{4}+2k\pi)] \quad k \in \mathbb{Z}$$

$$(4) \operatorname{Arg} z = \arctan \frac{1}{1} = -\frac{\pi}{4}, \quad \arg z = -\frac{\pi}{4} + 2k\pi, \quad k \in \mathbb{Z}$$

$$\therefore (1-i)^i = \sqrt{2} e^{i(\frac{\pi}{4}+2k\pi)i} = \sqrt{2} e^{\frac{\pi}{4}-2k\pi} \quad k \in \mathbb{Z}$$

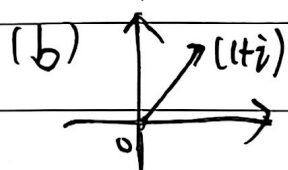
$$(1-i)^i = e^{i \ln(1-i)} = e^{i(\ln \sqrt{2}) + i(2k-\frac{1}{4})\pi} = e^{i \frac{1}{2} \ln \sqrt{2}} e^{(4k-1)\pi}$$

P80 1. (2)(3)



$$z = e^{i\theta} \quad \bar{z} = e^{-i\theta}$$

$$\therefore \int_C \bar{z} dz = \int_0^{2\pi} e^{-i\theta} i(e^{i\theta}) \cdot d\theta = i \int_0^{2\pi} d\theta = 2\pi i$$



$$z = x+iy$$

$$\int_C \bar{z} dz = \int_C (x-iy) d(x+iy) = \int_C x dx + y dy + i \int_C (x dy - y dx)$$

$$= \frac{1}{2} + \frac{1}{2} + i \int_0^1 0 dt = 1$$

$$\int_0^1 2t dt + i \int_0^1 0 \cdot dt = 1$$

$$+ i \int_C (x^2 - y^2) dy$$

$$(3) \int_C (x^2 - y^2) d(x+iy) = \int_C (x^2 - y^2) dx + i \int_C (x^2 - y^2) dy$$

$$x=0 \Rightarrow \text{LHS} = \int_0^1 -y^2 dy = -\frac{1}{3}$$